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Enhancing Early Detection of Parkinson's Disease Through Ensemble Learning and Nature-Inspired Feature Selection

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ABSTRACT

Parkinson's disease (PD) is a common central nervous system disorder that affects patients' speech abilities. Early detection of PD is essential for improving the treatment process. In the past decade, extensive research has been conducted on the early diagnosis of PD using artificial intelligence and machine learning methods. Feature selection and ensemble learning have shown to be effective strategies for designing diagnostic models. This study proposes an ensemble learning model combining support vector machine (SVM) and random forest (RF) algorithms to enhance the learning process. To reduce dimension and further improve the learning process, we employ nature-inspired meta-heuristic algorithms, such as binary particle swarm optimization (BPSO), binary butterfly optimization algorithm (BBOA), and binary whale optimization algorithm (BWOA) for feature selection. We utilized the standard PD dataset from the UCI database for our experiments. Our results demonstrate that the combination of BWOA-based feature selection and ensemble classifier achieves the highest performance, with an impressive accuracy of 97.43%. The proposed method effectively combines the strengths of BWOA for feature selection and ensemble learning for PD identification, resulting in exceptional accuracy and showcasing its potential for early diagnosis and treatment of the disease. The primary limitations of this study include its dependence on data quality, potential computational complexity, and the need for further validation across diverse populations.

1 | Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder that affects the central nervous system, specifically an area of the brain called the substantia nigra. Figure 1 provides a visual representation of the impact of PD on the substantia nigra [1]. In PD, there is a loss of dopamine-producing neurons in this region, resulting in reduced dopamine levels [2]. Dopamine is a critical neurotransmitter responsible for transmitting signals within the brain that control movement and coordination [3]. The severity and progression of symptoms vary among individuals. Currently,

there is no cure for PD, but medications, lifestyle modifications, and surgical interventions can help manage the symptoms and improve the quality of life. In many previous works, the issue of the importance of early diagnosis of this disease has been discussed [4–6]. Early detection of PD is crucial for several reasons:

- Better treatment outcomes: Early intervention can significantly slow the progression of the disease, which in turn can help maintain a better quality of life for a longer period. Certain medications, such as levodopa, are more effective in the early stages of PD.

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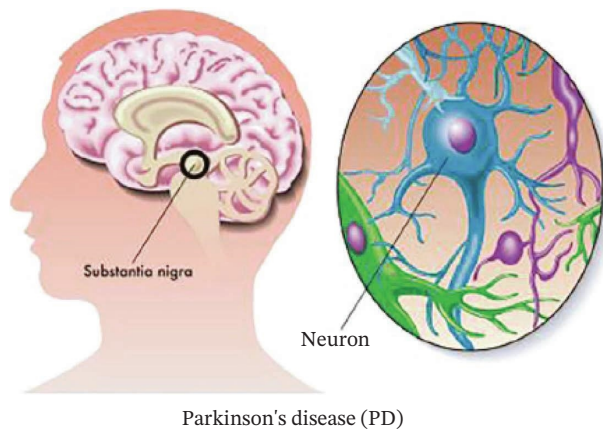


FIGURE 1 | The impact of PD on the substantia nigra [1].

- **Neuroprotective strategies:** Early detection allows for the potential use of neuroprotective strategies that could slow or halt the progression of PD. These strategies are generally more effective in the early stages of the disease when more dopamine neurons are still intact.
- **Psychological preparation:** Receiving an early diagnosis can provide individuals with more time to adjust to the emotional and psychological implications of having a neurodegenerative disease, seek support, and make plans for the future.

Machine learning has the potential to significantly improve early detection of PD [7–9]. Machine learning algorithms can analyze large datasets, including speech, gait, handwriting, and other biomarkers, to identify subtle patterns indicative of PD that might not be apparent to human observers. Machine learning approaches have demonstrated the potential for accurate classification of individuals with PD and those without, capitalizing on unique characteristics associated with the condition for early and precise detection.

Feature selection plays a significant role in improving the prediction accuracy of PD models [10]. By identifying the most relevant and informative features, feature selection helps to reduce the dimension of the data and enhance model interpretability. Several studies have demonstrated the effectiveness of various feature selection techniques in PD prediction [11–14]. Furthermore, the application of feature selection methods in conjunction with machine learning algorithms, such as decision tree (DT) [15], random forest (RF) [16], support vector machine (SVM) [17], K-nearest neighbors (KNN) [18], and artificial neural network (ANN) [19], has resulted in improved PD detection accuracy. This demonstrates the importance of selecting appropriate features and machine learning models for optimal PD prediction.

Ensemble learning has significantly impacted PD detection by improving the accuracy, robustness, and reliability of predictive models. This approach combines multiple learning algorithms, also known as base learners, to obtain better predictive performance than individual models [20]. Several studies have demonstrated the effectiveness of ensemble learning methods in PD detection [21–23]. Overall, ensemble learning has proven to be a powerful tool in PD detection, enabling researchers and clinicians to develop more accurate, reliable, and robust predictive models that can aid in early diagnosis and intervention.

To address the limitations of existing PD detection models, such as suboptimal accuracy and robustness, this study aims to enhance early diagnosis through the integration of machine learning and optimization techniques. Despite the success of various machine learning methods, there remains a need for more reliable and computationally efficient approaches. Therefore, we explore the application of three population-based optimization algorithms to optimize feature selection. These algorithms are chosen for their proven effectiveness in global search and convergence properties, which can help improve the predictive performance and robustness of PD detection models.

1.1 | Contributions

The key contributions of this paper can be outlined as follows:

- To present an ensemble learning model based on SVM and RF as a classifier to discriminate between normal individuals and patients.
- To apply population-based optimization algorithms, namely, binary particle swarm optimization (BPSO), binary butterfly optimization algorithm (BBOA), and binary whale optimization algorithm (BWOA) to extract the best features for PD diagnosis.

This study aims to assess the performance of the proposed methodology through the analysis of key metrics such as accuracy, sensitivity, specificity, selection rate, and other metrics. The results will be compared with the current state-of-the-art methods reported in the literature to demonstrate the superior performance and effectiveness of the proposed method in PD diagnosis.

This study introduces a novel approach that combines feature selection with the use of a classifier group, rather than relying on a single classifier. By integrating multiple classifiers into the feature selection process, the method leverages the strengths of different classifiers to improve overall classification performance and robustness. Unlike traditional methods that focus solely on feature subset quality, this approach embeds the classifier group directly into the fitness evaluation of the metaheuristic algorithm. This means that the selection process considers how well the feature subset performs across various classifiers, leading to more generalized and effective feature subsets that enhance classification accuracy across different models. The method considers two conflicting objectives, such as maximizing classification accuracy while minimizing feature subset size within a multiobjective fitness function. This multiobjective framework enables the algorithm to find a balanced trade-off between competing criteria, resulting in more practical and efficient solutions that are both accurate and parsimonious. This research advances the state-of-the-art by introducing a comprehensive and multifaceted optimization framework that simultaneously considers feature relevance, classifier diversity, and multiple performance objectives, leading to more robust and effective feature selection strategies in machine learning applications.

1.2 | Motivation

The current work serves as a strong driving force for continued exploration in the area of PD diagnosis. The demonstrated potential of integrating machine learning and optimization techniques to accurately detect the initial stages of PD and halt its

progression signifies a substantial advancement in the field of neuroscience. This innovative approach not only offers new avenues for early intervention but also contributes to a deeper understanding of the underlying mechanisms of this debilitating condition. The remarkable accuracy achieved by the proposed method positions it as a promising tool for early PD diagnosis. The potential implications of this research are significant, as it paves the way for further investigations to refine the model's accuracy and delve into additional learning techniques that can identify symptoms associated with the early stages of PD.

1.3 | Organization

The structure of this paper proceeds as follows:

- Section 2: A comprehensive review of the current state of research on diagnosing PD using deep learning, feature selection, ensemble classifier, and optimization techniques.
- Section 3: An in-depth description of the proposed methodology, which integrates the feature selection and ensemble classifier to enhance PD prediction.
- Section 4: A detailed account of the experimental setup, including the dataset used, evaluation metrics, and the results obtained when applying the proposed method. This section also provides a thorough discussion of the experimental results achieved.
- Section 5: A conclusion summarizing the key findings of the present study and offering insights into potential future research directions.

2 | Related Works

This section delves into the background of the recommended methodologies and highlights the advancements in our comprehension of learning models, which have been facilitated by prior research. These foundational studies have paved the way for the development of innovative approaches and contributed to the progression of knowledge in the field.

In a recent study conducted by the authors of [24], the authors investigated the application of machine learning classification algorithms for predicting PD using vocal disorder characteristics. The primary objective of the research was to enhance PD diagnosis by examining nonmotor symptoms, particularly vocal disorders, which affect approximately 90% of patients during the early stages of the disease. To achieve this, the study employed autoencoders to extract relevant features efficiently from vocal characteristic data, with the aim of identifying informative features that could improve the accuracy of PD diagnosis. The findings of the study demonstrated that the utilization of autoencoders for feature extraction, combined with logistic regression (LR) for classification, resulted in an impressive accuracy of 97.22%. In a related study [25], the authors explored the issue of dimensionality reduction for PD classification. The primary focus of the research was to evaluate the performance of genetic algorithm (GA) and BPSO in feature selection for PD diagnosis. To achieve this, the study employed both SVM and multilayer perceptron (MLP) for classification purposes. The experimental results revealed that the combination of an BPSO and an MLP yielded the highest detection accuracy of 89%.

In the study [26], the authors presented a method based on discrete wavelet transform to extract key movement features in PD. The study employed both SVM with linear kernel function and ANN for classification tasks. The results showed that the combination of discrete wavelet transform and ANN achieved the highest detection accuracy of 93.8%. In a relevant study [27], the authors emphasized the significance of medical data and the critical decisions derived from it, along with the exponential growth in data volume. These factors have driven researchers to develop feature selection approaches that identify the most pertinent data for addressing specific medical challenges. The paper explored two intelligent wrapper feature selection methods based on the coronavirus herd immunity optimizer (CHIO), with and without incorporating a greedy crossover (GC) operator strategy. The aim was to improve the exploration of the search space by CHIO. The experimental results demonstrated that the proposed approach, which combined CHIO and the GC operator strategy, achieved an accuracy of 79.03% using the KNN classifier.

In a notable study by the authors of [28], the researchers advanced the field of voice data analysis through two significant contributions. The first involved examining the integration of feature selection with deep neural networks (DNN) and assessing its effectiveness by contrasting its performance against conventional DNN models. The second contribution entailed the development of a novel ensemble model, which built upon recent research findings. The ensemble model demonstrated a 90% diagnosis accuracy, showcasing the potential benefits of incorporating feature selection techniques and capitalizing on recent research advancements in the development of high-performance classification models for voice data analysis. In a study by the authors of [29], a hybrid method integrating GA and SVM was proposed for PD diagnosis. The GA was applied for dimensionality reduction, while the SVM performed classification tasks. This approach contributes to ongoing research efforts in developing efficient diagnostic tools for PD by investigating the potential of combining metaheuristic optimization techniques with machine learning classifiers. The model showcased its effectiveness by achieving a 91.18% diagnosis accuracy, highlighting the significance of feature selection and classification in enhancing the performance of PD diagnostic tools.

In a relevant study by the authors of [30], the researchers proposed a unique approach that combined signal processing and resampling techniques, including oversampling, undersampling, and a hybrid combination of both. These resampling methods were integrated into well-established classifiers such as MLP and RF to improve classification performance. This research contributes to the continuous efforts in advancing classification techniques by exploring the potential advantages of combining resampling strategies with widely used machine learning algorithms. The proposed method demonstrated its effectiveness by achieving an accuracy of 93.81%. In a study by the authors of [31], a method was developed for the detection of PD from voice clips, utilizing a combination of filter feature selection, ensemble learning, and genetic selection techniques. The authors reported an accuracy of 91.53%.

2.1 | Comparison of Previous Methods

The reviewed studies [24–31] explored various aspects of PD diagnosis, with a primary emphasis on integrating feature selection techniques with machine learning algorithms to enhance diagnostic accuracy. Among these studies, the authors of [24] achieved a notable accuracy of 97.22% using a combination of autoencoders for feature extraction from vocal disorder characteristics and LR. This finding demonstrates the potential effectiveness of autoencoders in identifying informative features for early PD diagnosis through nonmotor symptoms. A comparative analysis of the previously discussed methods is presented in Table 1.

2.2 | Research Gaps

Based on the review of the studies [24–31], several research gaps can be identified in the field of PD prediction utilizing feature selection and ensemble classifiers:

- **Advanced ensemble methods:** While ensemble learning has been employed in some studies, more advanced ensemble methods and their combinations could be investigated to further enhance classification performance.
- **Advanced feature selection techniques:** Most studies have used traditional feature selection methods, while advanced techniques, such as deep learning-based methods or evolutionary algorithms, could potentially enhance the identification of informative features.
- **Robustness evaluation:** The robustness of ensemble classifiers and feature selection methods under different data distributions, noise levels, and real-world conditions needs to be further investigated.

Addressing these research gaps will contribute to the development of more accurate, robust, and interpretable prediction models based on feature selection and ensemble classifiers, ultimately improving decision-making and outcomes across various application domains.

2.3 | Research Evolution

The evolution of research in the field of PD detection and diagnosis has seen significant advancements over the years, driven by the continuous development of new technologies and methodologies. Below is a high-level overview of the key milestones and trends in this research area:

- **Early research:** Initial studies focused on identifying clinical symptoms and biomarkers associated with PD, which involved manual observation and clinical assessment of patients. This period laid the foundation for understanding the underlying causes and progression of the disease.
- **Introduction of machine learning:** With the advent of machine learning, researchers began exploring data-driven approaches for PD detection and diagnosis. Early applications of machine learning focused on developing classification models using clinical and neuroimaging data, demonstrating the potential of these methods in detecting PD more accurately and objectively.
- **Advanced feature selection techniques:** As data dimensionality and complexity increased, feature selection became an essential component of PD detection models. Researchers began investigating advanced techniques, such as filter, wrapper, and embedded methods, to identify the most relevant features for improved model performance.
- **Metaheuristic optimization:** In recent years, metaheuristic optimization algorithms, such as PSO and GA, have been employed to enhance feature selection and model optimization in PD detection. These methods have shown promising results in improving detection accuracy and feature selection efficiency.
- **Integration of nature-inspired and ensemble learning:** More recently, researchers have begun combining nature-inspired optimization techniques with ensemble learning strategies to further enhance the performance of PD detection models. These approaches aim to achieve a balance between exploration and exploitation in feature selection, leading to more accurate and robust detection systems.

The evolution of PD detection research has been marked by a gradual shift from traditional clinical assessment methods to advanced data-driven approaches. With the integration of machine learning, feature selection, metaheuristic optimization, and ensemble learning, researchers continue to push the boundaries of PD detection accuracy and efficiency, contributing to better patient outcomes and healthcare management.

3 | Methodology

In this section, a detection model based on feature selection and ensemble classifier is designed to diagnose PD. The proposed model mainly focuses on feature selection based on BPSO,

TABLE 1 | A comparative analysis of the previously discussed methods.

Ref	Feature selection	Techniques	Classifiers	Dataset	Accuracy (%)
[24]	✓	Autoencoders	RL	UCI	97.22
[25]	✓	GA/BPSO	SVM/MLP	UCI	89.00
[26]	✓	Discrete wavelet transforms	SVM/ANN	Sensor based	93.80
[27]	✓	CHIO-GC	KNN	UCI	79.03
[28]	✓	DNN	Ensemble classifier	UCI	90.00
[29]	✓	GA	SVM	UCI	91.18
[30]	×	—	ANN-MLP	UCI	93.81
[31]	✓	Quasiconstant feature	Ensemble classifier	UCI	91.53

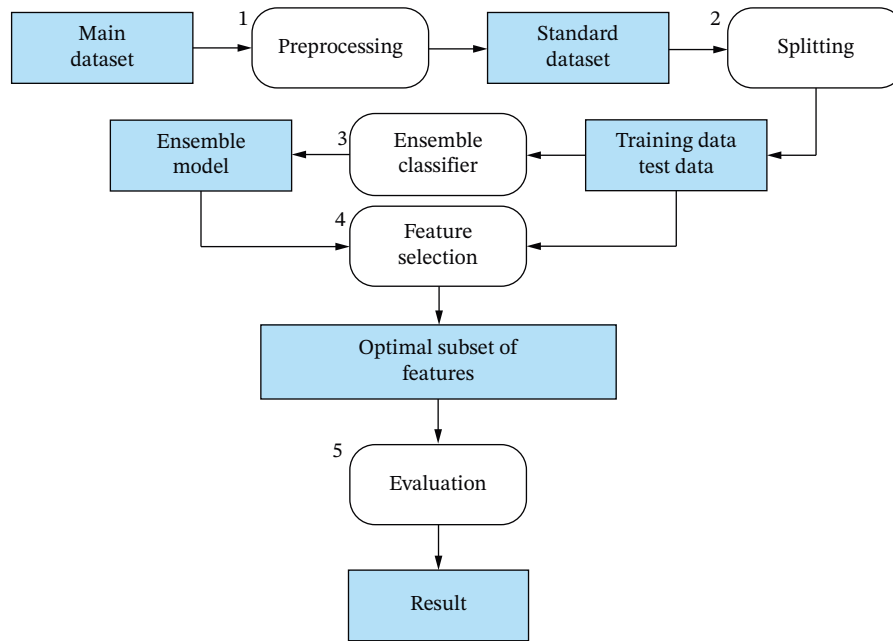


FIGURE 2 | Overview of the proposed model.

BBOA, and BWOA in PD. Figure 2 depicts an overview of the proposed model. The proposed model consists of five subsections, namely, preprocessing, splitting, ensemble classifier, feature selection, and evaluation.

3.1 | Preprocessing

Preprocessing techniques, such as normalization, play a vital role in improving the accuracy and efficiency of machine learning models for PD prediction [32]. These methods help transform raw data into a format that is more suitable for analysis, ensuring that the model can learn and generalize from the input data effectively. Normalization is a preprocessing technique used in machine learning to transform input data into a standard range or distribution. It ensures that all features have a similar scale, preventing bias towards features with larger absolute values and improving model convergence during training. In the context of data preprocessing, normalization typically involves rescaling feature values to fall within a specific range, usually between 0 and 1. This standardization process allows different features to be compared meaningfully and reduces the chances of a model being biased towards features with higher absolute values. Feature normalization is performed using Equation (1), where the Min/Max normalization technique standardizes the values of each feature to fall within the [0, 1] interval [33].

$$X_{\text{normal}} = \left(\frac{X - X_{\min}}{X_{\max} - X_{\min}} \right), \quad (1)$$

where X is the original feature value, $X_{\min}$ and $X_{\max}$ are the minimum and maximum of the feature, respectively, and X_{normal} is the normalized value.

As part of the preprocessing pipeline, all features were normalized using Min–Max normalization to map their values to the [0, 1] range, ensuring comparable scales across features. Following normalization, the dataset was split into 60% training and

40% testing subsets to address class imbalance and to allow reproducible performance evaluation.

3.2 | Splitting

The PD dataset is imbalanced due to the nature of the disease's prevalence and the data collection process. Imbalance in this context means that the number of samples representing PD patients (positive class) is typically much higher than the number of samples representing healthy individuals (negative class) [34]. The imbalance in PD datasets can present challenges in developing and evaluating machine learning models, as the models may become biased towards the majority class (PD patients), leading to poor performance in detecting healthy individuals accurately. Upon completing the imputation procedure, it became apparent that the dataset was imbalanced, comprising 147 instances of PD patients and 48 healthy controls, corresponding to 75% and 25% of the dataset, respectively. To address potential challenges in model performance, including underfitting or overfitting, the dataset was divided into a 60:40 train/test split, ensuring a more balanced distribution of samples during the training and testing phases.

3.3 | Ensemble Classifier

This section introduces an ensemble classifier that combines SVM and RF to improve the accuracy of PD detection. The primary aim of designing the ensemble classifier is to enhance decision accuracy by leveraging the strengths of multiple classification algorithms. The proposed ensemble classifier includes an initial set of 10 different classifiers, composed of 5 SVM expert systems and 5 RF expert systems. Table 2 details the properties of the 10 initial expert systems.

- SVM expert systems vary in their radial basis function (RBF) kernel hyperparameter, which influences the decision boundary between classes [35].

TABLE 2 | Properties of 10 initial expert systems.

Item	Name	Classifier	Parameter
1	C ₁	SVM	RBF = 1.1
2	C ₂	SVM	RBF = 1.2
3	C ₃	SVM	RBF = 1.3
4	C ₄	SVM	RBF = 1.4
5	C ₅	SVM	RBF = 1.5
6	C ₆	RF	NTree = 30
7	C ₇	RF	NTree = 50
8	C ₈	RF	NTree = 70
9	C ₉	RF	NTree = 90
10	C ₁₀	RF	NTree = 110

- RF expert systems differ in the number of DTs, which affects the overall complexity and predictive performance [36].

3.3.1 | Selection Step

Based on Table 2, an initial set of 10 classifiers is established. Subsequently, each classifier's accuracy on the PD dataset is calculated, and the classifier with the highest accuracy is identified as the best performer. To determine the selected classifiers, the accuracy of the remaining classifiers is compared to that of the best classifier. If the difference in accuracy is less than a predefined threshold, the classifier is considered suitable for selection. Algorithm 1 summarizes the classifier selection step, where "O" represents the number of selected classifiers. The threshold value (T) can be empirically adjusted to modify the number of selected classifiers. This allows the system designer to fine-tune the ensemble classifier based on the desired balance between accuracy and computational complexity.

3.3.2 | Voting Step

Once the classifiers are selected, a voting system is established to combine their predictions. For each test sample, each classifier casts a vote: if a classifier classifies the test sample as healthy, it outputs +1; if it classifies the sample as ill, it outputs -1. Consequently, multiple votes are generated for each test sample. The final vote for a test sample is computed using equation (2), which aggregates the votes from all selected classifiers. This voting system enables the ensemble classifier to leverage the collective predictive power of its individual components, ultimately enhancing the accuracy of PD detection.

$$\text{Final}_{\text{decision}} = \text{sgn}\left(\sum_{i=1}^O Y_i\right), \quad (2)$$

where Y_i represents the vote of the i th classifier for the test sample, and sgn is a standard mathematical function [37]. When the result of equation (2) is +1, it indicates that the ensemble classifier has classified the test sample as healthy. Conversely, a result of -1 signifies that the test sample has been classified as ill by the ensemble classifier. If the outcome of equation (2) is 0, it implies that the ensemble classifier could not determine the class of the test sample, resulting in a failed decision-making process. This procedure is repeated for all

ALGORITHM 1 | Classifier Selection Step.

Input:

Initial \ \ the Initial Set of Classifiers

T \ \ Threshold level

Output:

Optimal \ \ the Optimal set of classifiers

01. Initial = {C₁, C₂, ..., C₁₀}

02. Optimal = {}

03. Max_Accuracy = max (Accuracy (C₁), Accuracy (C₂), ..., Accuracy (C₁₀))

04. for $i = 1:10$

05. Diff = Max_Accuracy - Accuracy (C_i)

06. if Diff > T

07. Optimal = Optimal + C_i

08. end if

09. end for

10. O = |Optimal|

test samples, and the accuracy of the ensemble classifier is evaluated based on the collective results. By aggregating the votes from individual classifiers, the ensemble classifier aims to provide more accurate and reliable predictions for PD detection.

3.4 | Feature Selection

Feature selection plays a pivotal role in improving the accuracy and efficiency of machine learning models for PD diagnosis. The primary purpose of feature selection is to identify the most relevant features that contribute significantly to the prediction outcome while reducing data dimensionality and computational complexity. Feature selection helps address these challenges by identifying the most informative features that enable the model to distinguish between healthy individuals and those affected by PD effectively.

3.4.1 | Problem Description

Feature selection can be described as a problem that involves identifying the most relevant and informative features from a given dataset to improve the performance of machine learning models. Feature selection can be formulated as an optimization problem, where the goal is to find the optimal subset of features that maximizes model performance while minimizing the number of selected features [38].

3.4.2 | Coding Solutions

Binary coding is a technique used in optimization algorithms for feature selection, particularly in meta-heuristic algorithms such as BPSO, BBOA, and BWOA. In the context of feature selection, binary coding helps to represent the selected or discarded state of a feature as a binary value, either 0 or 1. In binary coding, each feature is assigned a binary variable, where

- 1 indicates that the feature is selected.
- 0 indicates that the feature is not selected.

By using binary coding, the feature selection problem is transformed into a binary optimization problem. Binary coding simplifies the implementation of meta-heuristic algorithms for feature selection by allowing the search process to focus on finding the most relevant features in a discrete binary space, rather than dealing with continuous values. This discrete representation helps maintain a balance between exploration and exploitation of the feature space, ensuring that the algorithm converges to an optimal feature subset that enhances the performance of machine learning models in tasks such as PD diagnosis.

3.4.3 | Fitness Function

In the context of PD diagnosis, a fitness function is used to evaluate the performance of a feature subset selected by an optimization algorithm. When considering two goals, such as classification accuracy and feature selection rate, a multi-objective fitness function can be formulated. A possible formulation of a multiobjective fitness function that combines accuracy and feature selection rate could be

$$\text{Fitness}(S) = \alpha * \text{Accuracy}(S) + (1 - \alpha) * (1 - \text{SelectionRate}(S)). \quad (3)$$

Here, S represents the selected feature subset, $\text{Accuracy}(S)$ denotes the classification accuracy achieved using the features in S , and $\text{SelectionRate}(S)$ represents the proportion of selected features in S compared to the total number of features. The parameter α is a weight that balances the importance of accuracy and feature selection rate, with values between 0 and 1. A higher α value gives more importance to accuracy, while a lower α value emphasizes the feature selection rate. By optimizing this multiobjective fitness function, the optimization algorithm can find a set of features that offer the best trade-off between accuracy and feature selection rate for PD diagnosis. These solutions can then be further analyzed to select the most suitable feature subset based on the specific requirements of the diagnosis task.

To further elucidate the fitness function's role in the context of PD diagnosis, it is crucial to delve deeper into the multiobjective optimization aspect and the impact of the weight parameter α . The multiobjective fitness function considers both classification accuracy and feature selection rate to identify an optimal subset of features for PD diagnosis. By combining these objectives, the algorithm strives to achieve a balance between high diagnostic accuracy and a reduced number of features, which can simplify the diagnostic process and improve its practical applicability. The weight parameter α plays a pivotal role in determining this balance. Higher values of α prioritize classification accuracy, leading the algorithm to favor feature subsets that yield better performance in detecting PD cases. Conversely, lower α values emphasize feature selection rate, encouraging the algorithm to find subsets with fewer features while still maintaining reasonable diagnostic accuracy. It is essential to carefully choose an appropriate value for α based on the specific requirements of the PD diagnosis task. In scenarios where diagnostic accuracy is of utmost importance, a higher α value would be suitable. On the

other hand, if simplicity and cost-effectiveness are primary concerns, a lower α value would be more appropriate, as it would promote the selection of fewer features while still ensuring a satisfactory level of accuracy.

3.4.4 | BPSO

The BPSO is a popular meta-heuristic algorithm adapted for feature selection in machine learning. The BPSO is particularly useful in binary optimization problems, making it suitable for selecting the most relevant features from a given dataset. In the context of feature selection, BPSO operates as follows:

1. We initialize a swarm of particles, where each particle represents a potential solution to the feature selection problem. In BPSO, particles are encoded as binary vectors, where each dimension corresponds to a feature and a binary value indicates whether the feature is selected or not.
2. We calculate the fitness of each particle using a fitness function that evaluates the classification performance and feature selection rate of the corresponding feature subset.
3. We update the personal best position of each particle if its current fitness is better than its previous best.
4. We update the global best position of the swarm if any particle has a better fitness than the current global best.
5. We update the velocity and position of each particle based on its personal best position, the global best position, and its current position. In BPSO, velocities are also binary vectors, and position updates involve applying an S-shaped transfer function [39] or a V-shaped transfer function [40] to the velocity values to obtain binary positions.
6. We repeat Steps 2–5 until a stopping criterion is met, such as reaching a maximum number of iterations. To ensure that the BPSO algorithm achieves an adequate balance between computational efficiency and solution quality, it is crucial to define an appropriate stopping criterion. In our implementation, we selected the number of iterations as the stopping condition for the algorithm. The BPSO iteratively searches for the optimal solution, updating particle positions and velocities based on cognitive and social components. The algorithm continues this process until it reaches a predefined maximum number of iterations, at which point it terminates. Selecting an appropriate number of iterations is essential to prevent excessive computational costs while still ensuring that the algorithm has sufficient time to converge to a satisfactory solution.

3.4.5 | BBOA

The BBOA is a metaheuristic optimization algorithm that has been adapted for feature selection tasks in machine learning. The algorithm is inspired by the foraging behavior of butterflies and operates in a binary search space, making it suitable for addressing feature selection problems. In the context of feature selection, BBOA works as follows:

1. We initialize a population of butterflies, where each butterfly represents a potential solution to the feature selection problem. The position of each butterfly is encoded as a binary vector, with each dimension corresponding to

a feature and a binary value indicating whether the feature is selected or not.

2. We calculate the fitness of each butterfly using a fitness function that evaluates the classification performance and feature selection rate of the corresponding feature subset.
3. We update the fragrance of each butterfly based on its fitness, where a higher fragrance indicates a better solution.
4. We perform the butterfly adjustment phase, where butterflies update their positions based on the best global solution, and the fragrance of other butterflies in the population. In BBOA, position updates involve applying an S-shaped transfer function [41] or a V-shaped transfer function [42] to continuous values to obtain binary positions.
5. We repeat Steps 2–4 until a stopping criterion is met, such as reaching a maximum number of iterations or achieving a satisfactory fitness level.

3.4.6 | BWOA

The BWOA is a metaheuristic optimization algorithm adapted for feature selection in machine learning. It is based on the whale optimization algorithm (WOA), which mimics the social behavior of humpback whales. BWOA operates in a binary search space, making it suitable for addressing feature selection problems. The BWOA algorithm can be applied to feature selection as follows:

1. We initialize a population of whales, where each whale represents a potential solution to the feature selection problem. The position of each whale is encoded as a binary vector, with each dimension corresponding to a feature and a binary value indicating whether the feature is selected or not.
2. We calculate the fitness of each whale using a fitness function that evaluates the classification performance and feature selection rate of the corresponding feature subset.
3. We perform the encircling prey, bubble-net attacking, and search for prey phases of the WOA algorithm to update the positions of the whales in the population. In BWOA, position updates involve applying an S-shaped transfer function [43] or a V-shaped transfer function [44] to continuous values to obtain binary positions.
4. We repeat steps 2–3 until a stopping criterion is met, such as reaching a maximum number of iterations or achieving a satisfactory fitness level.

3.4.7 | Binary Transfer Function

Binary transfer functions, such as S-shaped and V-shaped functions, play a crucial role in feature selection, particularly when used in binary optimization algorithms. These functions serve as a bridge between the continuous search space explored by optimization algorithms and the binary representation required for feature selection problems.

3.4.7.1 | S-Shaped Transfer Function. Let us consider the k th dimension of the i th solution at the t th iteration before the updating process, denoted as $X_i^k(t)$. After the updating process, this value is transformed into the k th dimension of the i th solution at the t th iteration, represented by $F_i^k(t)$. The transformation is achieved using an S-shaped transfer function, which can be calculated using the following equation:

$$S(F_i^k(t)) = \frac{1}{1 + e^{-F_i^k(t)}}. \quad (4)$$

The S-shaped transfer function is employed to convert the updated solution positions back into binary format. To accomplish this, a random number in the range of $[0, 1]$ is generated. Equation (5) is then utilized to determine whether the updated positions should be transformed into binary values, thereby ensuring that the feature selection process adheres to binary constraints.

$$X_i^k(t+1) = \begin{cases} 0, & \text{rand} < S(F_i^k(t)), \\ 1, & \text{rand} \geq S(F_i^k(t)). \end{cases} \quad (5)$$

3.4.7.2 | V-Shaped Transfer Function. We denote the k th dimension of the i th solution at the t th iteration before the updating process as $X_i^k(t)$. Upon completion of the updating process, this value is converted into the k th dimension of the i th solution at the t th iteration, symbolized by $F_i^k(t)$. The conversion employs a V-shaped transfer function, which can be computed using the following equation:

$$V(F_i^k(t)) = \left| \text{erf} \left(\frac{\sqrt{\pi}}{2} F_i^k(t) \right) \right|. \quad (6)$$

The V-shaped transfer function is utilized to transform solution positions back into binary format. The process begins with the generation of a random number in the range of $0-1$. Equation (7) is then used to decide whether the positions should be converted to binary values, thereby maintaining binary constraints in the feature selection process.

$$X_i^k(t+1) = \begin{cases} X_i^k(t), & \text{rand} < V(F_i^k(t)), \\ \sim X_i^k(t), & \text{rand} \geq V(F_i^k(t)). \end{cases} \quad (7)$$

3.4.7.3 | Hybrid Transfer Function. A novel hybrid transfer function is introduced in this section, which combines the S-shaped and V-shaped transfer functions. The hybrid function leverages the strengths of both transfer functions. We consider $X_i'(t+1)$ as the i th solution at iteration $t+1$, which is transformed into binary format using the S-shaped function. Similarly, let $X_i''(t+1)$ represent the i th solution at iteration $t+1$, modified by the V-shaped function. The proposed hybrid transfer function is computed using equation (8), where each solution is converted into binary format based on both S-shaped and V-shaped functions. The decision to use either function is made by comparing their respective fitness values.

$$X_i(t+1) = \begin{cases} X'_i(t+1), \text{fitness}(X'_i(t+1)) \geq \text{fitness}(X''_i(t+1)), \\ X''_i(t+1), \text{fitness}(X'_i(t+1)) < \text{fitness}(X''_i(t+1)). \end{cases} \quad (8)$$

3.5 | Evaluation

In the evaluation stage of our proposed PD prediction model, it is crucial to assess the performance and effectiveness of the ensemble classifier in conjunction with the selected features. The evaluation stage provides a comprehensive assessment of your PD prediction model's performance, feature importance, and statistical significance.

4 | Experimentation

The proposed methodology in this study is simulated using MATLAB Version 2017. The detailed description is provided in Appendix A. This section presents and discusses the simulation results obtained from the experiments. The initial parameter values for the BPSO, BBOA, and BWOA algorithms, listed in Table 3, are maintained consistently across all simulations to ensure reliable and comparable results. In an effort to establish the optimal initial parameter values for the metaheuristic algorithm, numerous simulations have been conducted using a range of initial values. This systematic approach ensures the selection of the most suitable initial values, ultimately contributing to the algorithm's performance and efficacy in solving the problem.

4.1 | Dataset

This paper utilizes a publicly available dataset from the University of Oxford repository, developed in collaboration

with the National Center for Voice. Initially intended for voice disorder research, the dataset includes voice recordings from 31 participants: 23 individuals diagnosed with PD and 8 healthy samples. Among the PD group, there are 16 males and 7 females, while the healthy group consists of 3 males and 5 females. The dataset contains a total of 195 voice recordings, each lasting 36 s and captured in a sound-controlled environment. To facilitate understanding, the dataset's "status" column assigns a value of "1" to individuals with PD and "0" to healthy samples. This binary classification allows for a clear distinction between the two groups and aids in analyzing the effectiveness of the proposed diagnostic methodology. Table 4 provides a comprehensive overview of the various voice measures present in the UCI dataset [45]. These measures serve as essential indicators of vocal characteristics, enabling an in-depth analysis of individuals with PD and healthy controls, ultimately aiding the development of accurate diagnostic methodologies.

This study utilizes a publicly available voice dataset from the University of Oxford repository, developed in collaboration with the National Center for Voice. The dataset comprises voice recordings from 31 participants, including 23 individuals diagnosed with PD and 8 healthy controls. Each recording lasts approximately 36 s and was captured in a sound-controlled environment. The dataset includes

TABLE 3 | Initial values for BPSO, BBOA, and BWOA parameters.

No.	Algorithm	Parameter	Value
1	BWOA	Whale length	22
2		Population size	300
3		Max iteration	250
4		(P) Control parameter	Randomly
5		Weight of fitness 1	0.6
6		Weight of fitness 2	0.4
7	BBOA	Butterfly length	22
8		Population size	300
9		Max iteration	250
10		(P) Control parameter	0.6
11		Weight of fitness 1	0.6
12		Weight of fitness 2	0.4
13		Sensory modality	0.01
14		Power exponent	0.1 to 0.3
15	BPSO	Particle length	22
16		Population size	300
17		Max iteration	250
18		Weight of fitness 1	0.6
19		Weight of fitness 2	0.4
20		Inertia weight	0.7
21		C ₁	2.5
22		C ₂	1.6

TABLE 4 | Features of the PD dataset.

No.	Index	Feature	Description
1	F01	MDVP:F0 (Hz)	The average vocal fundamental frequency
2	F02	MDVP:Fhi (Hz)	The maximum vocal fundamental frequency
3	F03	MDVP:Flo (Hz)	The minimum vocal fundamental frequency
4	F04	MDVP:Jitter (%)	The measure of variation in fundamental frequency
5	F05	MDVP:Jitter (Abs)	The measure of variation in fundamental frequency
6	F06	MDVP:RAP	The measure of variation in fundamental frequency
7	F07	MDVP:PPQ	The measure of variation in fundamental frequency
8	F08	Jitter:DDP	The measure of variation in fundamental frequency
9	F09	MDVP:Shimmer	The measure of variation in amplitude
10	F10	MDVP:Shimmer (dB)	The measure of variation in amplitude
11	F11	Shimmer:APQ3	The measure of variation in amplitude
12	F12	Shimmer:APQ5	The measure of variation in amplitude
13	F13	MDVP:APQ	The measure of variation in amplitude
14	F14	Shimmer:DDA	The measure of variation in amplitude
15	F15	NHR	The measure of the ratio of noise to tonal components
16	F16	HNR	The measure of the ratio of noise to tonal components
17	F17	RPDE	The nonlinear dynamical complexity measure
18	F18	D2	The nonlinear dynamical complexity measure
19	F19	DFA	The signal fractal scaling exponent
20	F20	Spread1	The nonlinear measure of fundamental frequency variation
21	F21	Spread2	The nonlinear measure of fundamental frequency variation
22	F22	PPE	The nonlinear measure of fundamental frequency variation

various voice measures related to vocal characteristics, which are used as features for classification. The recordings were originally intended for voice disorder research; however, they provide valuable biomarkers for PD detection. Details such as sampling frequency are not explicitly specified in the dataset documentation. The “status” column indicates whether the participant has PD or healthy, facilitating binary classification. This dataset supports the development of diagnostic models aimed at early PD detection based on voice features.

The features extracted from the voice recordings are based on established acoustic and vocal characteristics known to be indicative of PD. These features include measures such as fundamental frequency, jitter, shimmer, and harmonic-to-noise ratio, among others.

It should be noted that the original dataset documentation does not explicitly specify the sampling frequency of the voice recordings. All voice features were extracted from 36-s recordings captured in a sound-controlled environment, and we applied a consistent preprocessing pipeline to ensure comparability of samples.

4.2 | Performance Metrics

To evaluate the performance of our model, we have employed different evaluation metrics, namely, accuracy (ACC), sensitivity (SEN), specificity (SPE), precision (PRE), F1-score, and geometric mean (GM). These metrics are defined in equations (9)–(14) as follows:

$$\text{Accuracy (ACC)} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (9)$$

$$\text{Sensitivity (SEN)} = \frac{TP}{TP + FN}, \quad (10)$$

$$\text{Specificity (SPE)} = \frac{TN}{TN + FP}, \quad (11)$$

$$\text{Precision (PRE)} = \frac{TP}{TP + FP}, \quad (12)$$

$$F1 - \text{score} = 2 \times \frac{PRE \times SEN}{PRE + SEN}, \quad (13)$$

$$\text{Geometric mean (GM)} = \sqrt{PRE \times SEN}, \quad (14)$$

where TP, FP, TN, and FN refer, respectively, to true positive, false positive, true negative, and false negative. The confusion matrix in Table 5 illustrates the performance of the learning model, enabling the calculation of various evaluation metrics.

TABLE 5 | The confusion matrix.

Confusion matrix		Actual	
		Positive	Negative
Predicted	Positive	TP	FP
	Negative	FN	TN

4.3 | Training and Testing Datasets

This groundbreaking study examines the potential of voice samples as a noninvasive diagnostic tool for identifying PD. Leveraging the UCI dataset, which comprises 195 voice recordings from 147 PD patients and 48 healthy controls, the research explores the efficacy of various machine learning algorithms for PD detection. To ensure a comprehensive assessment, the dataset is methodically divided into two subsets: 60% allocated for training and the remaining 40% designated for testing and validation purposes. Table 6 presents a detailed overview of the information pertaining to the training and testing datasets.

4.4 | Results

The evaluation results of classifiers without feature selection are presented in Table 7. Among the 10 primary classifiers, the RF classification with 90 trees achieved the highest accuracy, reaching 0.8589. To construct the ensemble classifier, C_9 was selected due to its superior performance. Table 8 illustrates the evaluation results of the ensemble classifier without feature selection. As indicated by the results, an accuracy level of 0.8974 was attained by setting the threshold to 0.035. Consequently, EC_3 was chosen as the optimal ensemble classifier for implementation within the binary optimization algorithms. Additional results are available in the Supporting section (Supporting Table S1–S7).

Table 9 illustrates the results of the proposed method when the feature selection problem is addressed using the BPSO and the ensemble classifier EC_3 . The comparison of the performance between the hybrid transfer function, the S-shaped transfer function, and the V-shaped transfer function suggests that the hybrid transfer function yields superior results in this context. Table 10 presents the results of the proposed method when the feature selection issue is tackled using the BBOA and the ensemble classifier EC_3 . A comparison of the performance between the hybrid transfer function, S-shaped transfer function, and V-shaped transfer function indicates that the hybrid transfer function outperforms the others in this context. Table 11 showcases the results of the proposed method when addressing the feature selection problem using the BWOA and the ensemble classifier EC_3 . The comparison of performance among the hybrid transfer function, S-shaped transfer function, and V-shaped transfer function reveals that the hybrid transfer function outperforms the others in this context.

Figures 3, 4, and 5 illustrate a comparative analysis of binary metaheuristic algorithms employing the S-shaped transfer function, the V-shaped transfer function, and the hybrid transfer function. The results obtained from this comparison indicate that the BWOA demonstrates superior performance when compared to the other two algorithms. This highlights the effectiveness of

TABLE 6 | The detailed overview of the information pertaining to the training and testing datasets.

Class	Training	Testing	Total
Healthy	21	27	48
Patient	96	51	147
Total	117	78	195

TABLE 7 | Evaluation results of classifiers without feature selection.

No.	Classifier	ACC
1	C_1	0.8205
2	C_2	0.8333
3	C_3	0.8333
4	C_4	0.8461
5	C_5	0.8461
6	C_6	0.7948
7	C_7	0.8333
8	C_8	0.8461
9	C_9	0.8589
10	C_{10}	0.8461

the BWOA in optimizing the feature selection process within the context of binary meta-heuristic algorithms. In conclusion, the BWOA emerges as the top-performing algorithm among those evaluated, showcasing its potential for achieving optimal results in feature selection tasks. Table 12 presents the fitness function results for the best solutions acquired from the various algorithms evaluated in this study. The fitness function serves as an indicator of each algorithm's performance in optimizing feature selection.

As demonstrated in Table 12, the BWOA achieves the highest fitness function result, outperforming both the BPSO and the BBOA. This outcome further emphasizes the effectiveness of BWOA in identifying optimal solutions for feature selection tasks. Table 13 presents the features chosen by the BWOA based on hybrid transfer function as the optimal subset for achieving high classification accuracy. The algorithm identified features 1, 4, 5, 11, 12, 16, 17, 19, 21, and 22 as the most relevant and informative for the given task, resulting in an impressive 97.43% accuracy. The selected features capture essential aspects of speech and voice production, including fundamental frequency, pitch variation, amplitude modulation, and vocal fold dynamics, which have been found to be significant indicators for identifying PD. The high accuracy achieved by this feature subset emphasizes the potential of the BWOA in identifying the most relevant features for PD classification tasks.

The combination of the 10 selected features demonstrated superior performance in distinguishing Parkinson's patients from healthy controls. These features are known to capture critical vocal impairments associated with PD, such as variations in pitch, amplitude, and voice stability, which are characteristic symptoms of the disease. The selected features effectively encapsulate the key acoustic markers that differentiate PD-affected voices from normal voices, thereby enabling accurate classification. This optimal feature subset enhances the model's

TABLE 8 | Evaluation results of ensemble classifiers without feature selection.

Name	T	Classifiers	ACC
EC_1	0.015	$C_9-C_{10}-C_8-C_4-C_5$	0.8846
EC_2	0.025	$C_9-C_{10}-C_8-C_4-C_5$	0.8846
EC_3	0.035	$C_9-C_{10}-C_2-C_3-C_7-C_8-C_4-C_5$	0.8974

Note: The bold values indicate the best results among the compared methods.

TABLE 9 | The results of the proposed method based on BPSO and ensemble classifier EC₃.

Metric	BPSO		
	S-shaped transfer function	V-shaped transfer function	Hybrid transfer function
ACC	0.8717	0.8974	0.9230
SEN	0.8823	0.9019	0.9215
SPE	0.8518	0.8888	0.9259
PRE	0.9183	0.9387	0.9591
F1-score	0.8999	0.9199	0.9399
GM	0.9001	0.9201	0.9401

Note: The bold values indicate the best results among the compared methods.

TABLE 10 | The results of the proposed method based on BBOA and ensemble classifier EC₃.

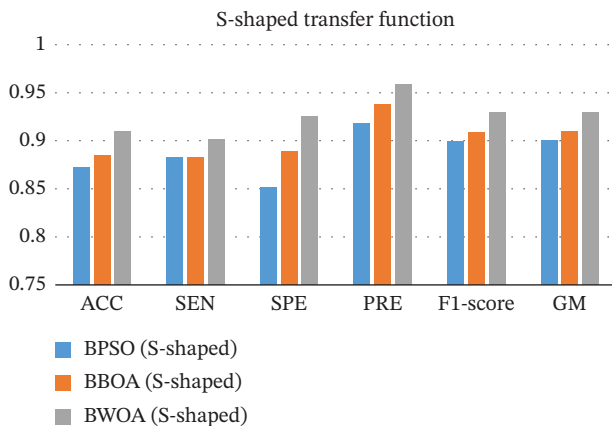
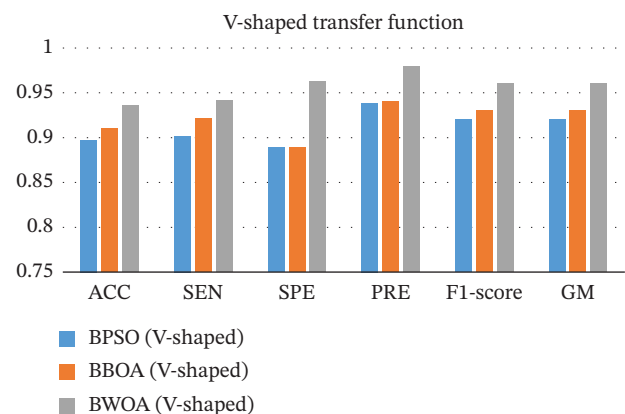
Metric	BBOA		
	S-shaped transfer function	V-shaped transfer function	Hybrid transfer function
ACC	0.8846	0.9102	0.9358
SEN	0.8823	0.9215	0.9411
SPE	0.8888	0.8888	0.9259
PRE	0.9375	0.9400	0.9600
F1-score	0.9090	0.9306	0.9504
GM	0.9094	0.9307	0.9505

Note: The bold values indicate the best results among the compared methods.

TABLE 11 | The results of the proposed method based on BWOA and ensemble classifier EC₃.

Metric	BWOA		
	S-shaped transfer function	V-shaped transfer function	Hybrid transfer function
ACC	0.9102	0.9358	0.9743
SEN	0.9019	0.9411	0.9803
SPE	0.9259	0.9629	0.9629
PRE	0.9583	0.9795	0.9803
F1-score	0.9292	0.9599	0.9803
GM	0.9296	0.9601	0.9803

Note: The bold values indicate the best results among the compared methods.

**FIGURE 3** | Comparative analysis of binary metaheuristic algorithms using S-shaped transfer function.**FIGURE 4** | Comparative analysis of binary metaheuristic algorithms using V-shaped transfer function.

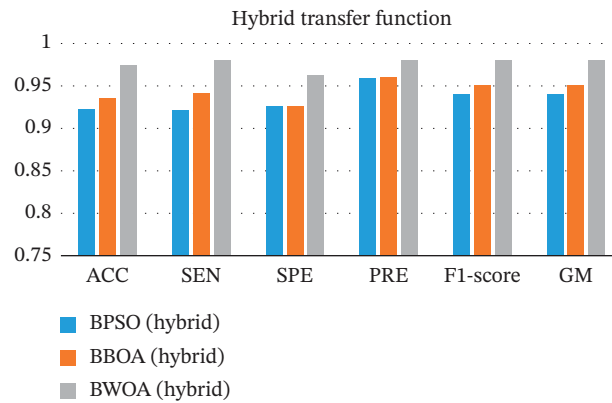


FIGURE 5 | Comparative analysis of binary metaheuristic algorithms using hybrid transfer function.

TABLE 12 | The fitness function results.

Algorithm	Transfer function	Accuracy (Fit1)	Selection rate (Fit2)	Fitness
BPSO	S-shaped	0.8717	0.6363	0.66850
	V-shaped	0.8974	0.5909	0.70208
	Hybrid	0.9230	0.5454	0.73564
BBOA	S-shaped	0.8846	0.6363	0.67624
	V-shaped	0.9102	0.5454	0.72796
	Hybrid	0.9358	0.5000	0.76148
BWOA	S-shaped	0.9102	0.5454	0.72796
	V-shaped	0.9358	0.4545	0.77968
	Hybrid	0.9743	0.4545	0.80278

TABLE 13 | The features chosen by the BWOA.

Algorithm	Transfer function	Feature selected
BWOA	S-shaped	F1/F5/F6/F9/F10/F11/F12/F16/F17/F19/F20/F21
	V-shaped	F4/F5/F6/F7/F8/F9/F11/F12/F17/F19
	Hybrid	F1/F4/F5/F11/F12/F16/F17/F19/F21/F22

Note: The bold values indicate the best results among the compared methods.

discriminative power by focusing on the most relevant vocal characteristics related to the pathology.

Table 14 presents a comprehensive comparison of classification accuracy achieved by different combinations of metaheuristic algorithms with various classifiers, highlighting the effectiveness of each approach. Notably, the combination of BWOA with an ensemble classifier yields the highest accuracy of 0.9743, demonstrating superior performance among all tested configurations. This suggests that leveraging the ensemble classifier alongside BWOA significantly enhances the model's ability to generalize and accurately classify data. The results also indicate that BWOA combined with individual classifiers like SVM and RF perform well, with accuracies of 0.9514 and 0.9487, respectively, underscoring the robustness of BWOA in optimizing feature selection or parameter tuning when paired with different classifiers. In contrast, BBOA and BPSO show comparatively lower accuracies across all classifiers, with the ensemble classifier again outperforming individual classifiers within these algorithms. This highlights the potential advantage of ensemble strategies in boosting classification performance. Overall, the findings

emphasize the importance of the choice of metaheuristic algorithm and classifier combination. The superior performance of BWOA with an ensemble classifier suggests that this hybrid approach effectively captures data patterns, leading to higher accuracy. Consequently, employing such integrated strategies can be highly beneficial in developing more accurate and robust classification models, particularly in complex or high-dimensional datasets.

The statistical analysis in Table 15 compares the performance of the proposed method (BWOA + ensemble classifier) with 8 other methods for feature selection in PD datasets. The first column lists the 8 methods that are being compared with the proposed. The second column shows the total number of comparisons made between the proposed method and each of the other methods. The third column indicates how many times the proposed method outperformed the other methods in terms of regarding the detection accuracy. The last column presents the p-level, which is a measure of statistical significance. In this case, a p-level of 1 suggests that the difference in performance between the proposed method and the other methods is statistically

TABLE 14 | Review of classification performance in the model.

No.	Best model	ACC
1	BWOA + ensemble classifier	0.9743
2	BWOA + SVM	0.9514
3	BWOA + RF	0.9487
4	BBOA + ensemble classifier	0.9358
5	BBOA + SVM	0.9211
6	BBOA + RF	0.9106
7	BPSO + ensemble classifier	0.9230
8	BPSO + SVM	0.9145
9	BPSO + RF	0.9121

significant, favoring the proposed method. The statistical analysis presented in Table 15 provides strong evidence that the proposed method outperforms the other 8 methods with regard to the detection accuracy. The consistent p-level of 1 across all comparisons supports the claim that the proposed method is statistically superior in identifying relevant features for PD classification.

4.5 | Comparison

Table 16 compares the performance of the proposed method with other existing methods. The results demonstrate that the proposed method, which combines the BWOA with an ensemble classifier, outperforms other techniques by achieving the highest accuracy in identifying PD. As seen in Table 16, the proposed method combines the strengths of BWOA for feature selection and an ensemble classifier for PD identification, resulting in a remarkable 97.43% accuracy. This highlights the effectiveness of the proposed approach in accurately detecting PD using speech features, surpassing other methods that utilize different optimization algorithms and classifiers. In order to attain the highest detection accuracy, a careful selection of features was performed from the available set of 22 features. The feature selection process, which plays a crucial role in enhancing the performance of the classification algorithm, resulted in identifying 10 highly relevant and informative features for the task of PD identification. These 10 features were chosen based on their significance and

TABLE 15 | The statistical analysis using Wilcoxon signed-rank test.

Methods	Number of comparisons runs	Number of wins	p-level
BWOA + SVM	15	14	1
BWOA + RF	15	14	1
BBOA + ensemble classifier	15	14	1
BBOA + SVM	15	15	1
BBOA + RF	15	15	1
BPSO + ensemble classifier	15	15	1
BPSO + SVM	15	15	1
BPSO + RF	15	15	1

TABLE 16 | The comparison of the proposed method with other methods.

Ref	Best model	ACC
[24]	Autoencoders + RL	0.9722
[25]	BPSO + MLP	0.8900
[27]	CHIO-GC + KNN	0.7903
[28]	DNN + ensemble classifier	0.9000
[29]	GA + SVM	0.9118
[30]	ANN-MLP	0.9381
[31]	Quasiconstant feature + ensemble classifier	0.9153
Proposed method	BWOA + ensemble classifier	0.9743

contribution to accurately distinguishing between PD patients and healthy controls. By selecting an optimal subset of features, the proposed method was able to increase the detection accuracy while reducing the computational complexity associated with processing all 22 features. This illustrates the importance of efficient feature selection in maximizing classification performance for PD identification using speech data. While the proposed ensemble learning approach combined with nature-inspired feature selection demonstrates promising results for early Parkinson's disease detection, it also has certain limitations. The effectiveness of the method heavily depends on the quality and representativeness of the available data, and variations in data collection protocols can impact model performance. Additionally, the computational complexity associated with the ensemble and the optimization processes may lead to increased training time, especially with large datasets or high-dimensional feature spaces. The reliance on heuristic or metaheuristic algorithms for feature selection may also result in suboptimal feature subsets if not properly tuned, potentially affecting classification accuracy. Furthermore, the model's generalizability to different populations or clinical settings requires further validation, as overfitting to specific datasets could limit its broader applicability.

5 | Conclusion

This study presented a novel ensemble classifier-based detection model for PD prediction, focusing on feature selection utilizing BPSO, BBOA, and BWOA. The proposed model's unique design combines an ensemble of SVM and RF expert systems to achieve enhanced classification performance. By integrating these elements into a comprehensive framework, our approach demonstrated significant improvements in diagnostic accuracy compared to existing techniques. The feature selection process, driven by the BPSO, BBOA, and BWOA algorithms, played a crucial role in identifying the most relevant and informative features for accurate PD prediction. The optimization techniques enabled the selection of an optimal subset of features, resulting in a more efficient and robust model. Our findings emphasize the importance of advanced feature selection methods and ensemble learning in developing effective diagnostic tools for PD. The proposed model's success in achieving high accuracy, sensitivity, and specificity paves the way for further research in this domain,

ultimately leading to improved patient care and early intervention.

The findings of this research can benefit various stakeholders, including clinicians, researchers, and patients. Clinicians can leverage the proposed ensemble learning and feature selection methods to improve the early diagnosis of PD, enabling timely intervention and personalized treatment plans. Researchers can utilize the developed framework as a foundation for further exploration into biomarker identification and the development of more accurate diagnostic tools. Patients stand to benefit from earlier detection, potentially leading to better management of the disease and improved quality of life. Additionally, healthcare institutions and policymakers can incorporate these advancements into screening programs, optimizing resource allocation and enhancing overall healthcare outcomes related to neurodegenerative diseases.

The proposed ensemble classifier-based detection model for PD prediction exhibits several advantages over existing methods:

- **Feature selection:** The usage of advanced feature selection techniques (BPSO, BBOA, and BWOA) enables the identification of the most relevant and informative features from the available dataset. This not only improves the model's accuracy but also reduces the computational complexity associated with processing a large number of features.
- **Ensemble classifier:** The proposed model combines SVM and RF expert systems, taking advantage of their complementary strengths to achieve enhanced classification performance. This ensemble approach results in more accurate and robust predictions compared to using a single classifier.
- **Higher accuracy:** The proposed model demonstrates superior accuracy, sensitivity, and specificity in detecting PD, surpassing the performance of existing methods. This increased accuracy enables earlier diagnosis and intervention, ultimately leading to better patient outcomes.
- **Adaptability:** The proposed model's framework is flexible and can incorporate additional optimization algorithms or classifiers to further improve its performance. Additionally, it can be applied to various datasets, showcasing its potential for generalization and real-world clinical applications.

While the proposed ensemble classifier-based detection model for PD prediction presents several advantages, there are a few potential limitations that should be considered:

- **Computational complexity:** The incorporation of feature selection algorithms (BPSO, BBOA, and BWOA) and the ensemble classifier (SVM and RF) may result in higher computational complexity compared to simpler models. This could increase the required processing time and resources, which might be a concern for real-time or resource-constrained applications.
- **Parameter tuning:** Optimizing the performance of the proposed model may require careful tuning of various parameters associated with the feature selection algorithms and the ensemble classifier. This process can be time-consuming and might necessitate expert knowledge in optimization techniques and machine learning.

Future work related to the proposed ensemble classifier-based detection model for PD prediction can focus on the following aspects:

- **Additional optimization algorithms and classifiers:** We explore the potential of incorporating additional feature selection techniques and machine learning classifiers into the proposed model. This could lead to further improvements in predictive performance and enhance the model's adaptability to different datasets and clinical settings.
- **Fine-tuning and hyperparameter optimization:** We investigate advanced optimization strategies for fine-tuning the parameters of the existing feature selection algorithms and ensemble classifier components. This could help maximize the model's performance and reduce computational complexity.
- **Real-time implementation:** We develop a real-time implementation of the proposed model to facilitate its integration into clinical decision support systems or wearable devices for continuous monitoring of patients at risk of developing PD.

Author Contributions

Seyyed Ahmad Hashemi: conceptualization, methodology, software, data curation, formal analysis, and writing and original draft.

Faraein Aeni: methodology, validation, and writing.

Homayun Motameni and Behnam Barzegar: validation, review, editing, and supervision.

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Disclosure

All authors have read and approved the final version of the manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data used in this study are publicly available from the UCI Machine Learning Repository. The dataset can be accessed at <https://archive.ics.uci.edu/dataset/174/parkinsons>. No new data were created in this study.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Supporting Information.** The detailed results of this study are provided in the Supporting section and are presented in > Tables S1–S7.

Appendix A: Code Availability

The complete MATLAB implementation of the proposed algorithm, along with example scripts to reproduce the experimental results, is publicly available at <https://github.com/Ahmad-cide/my-algorithm-implementation>.